**SAVEETHA INSTITUTE OF MEDICAL AND TECHNICAL SCIENCES**

**SIMATS ENGINEERING**

**SUB CODE & NAME: CSA17- ARTIFICIAL INTELLIGENCE**

**Exp 2: Write the python program to solve 8-Queen problem**

**CODE:**

```
N = 8
board = [-1] * N

def is_safe(row, col):
    for r in range(row):
        if board[r] == col:
            return False
        if abs(board[r] - col) == abs(r - row):
            return False
    return True

def solve(row):
    if row == N:
        return True

    for col in range(N):
        if is_safe(row, col):
            board[row] = col
            if solve(row + 1):
                return True
            board[row] = -1
```

```
    return False

if solve(0):
    print("One solution to 8-Queen problem:")
    for r in range(N):
        print(" ".join(
            "Q" if board[r] == c else "."
            for c in range(N)
        ))
else:
    print("No solution found")
```

**OUTPUT:**

